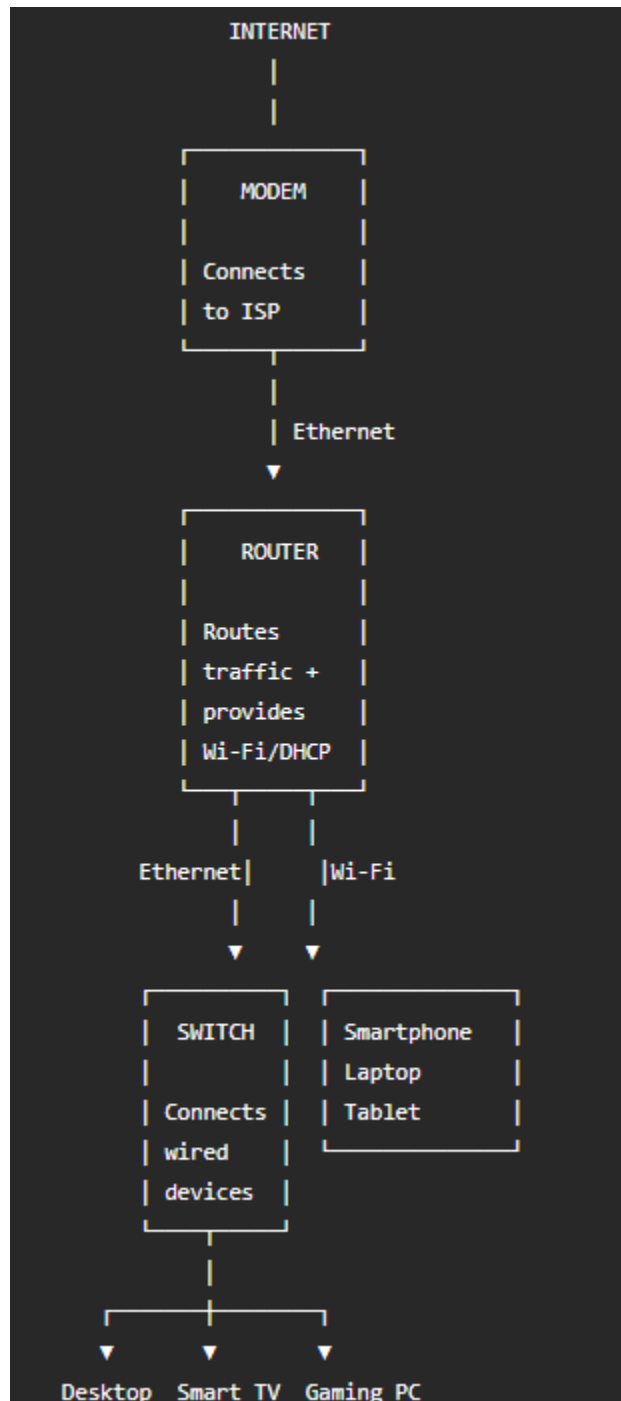


Task 1: Home Network Diagram



Role of Each Device

Modem:

Connects the home network to the ISP's Internet connection and converts the ISP's access signal into a form the local network can use.

Router:

Connects different networks, routes traffic between the home LAN and the Internet, and commonly provides services such as DHCP and Wi-Fi.

Switch:

Connects multiple wired devices within the same local network and forwards Ethernet frames to the appropriate device.

Task 2: Check Connected Devices on the Router

Steps

1. Connect my computer or phone to the home Wi-Fi.
2. Open a web browser.
3. Find the Default Gateway using:
 - ipconfig
4. Look for the Default Gateway, for example:
 - Default Gateway : 192.168.1.1
5. Enter that address into the browser.
6. Log in using the router's administrator credentials.
7. Open a section such as:
 - Connected Devices
 - Attached Devices
 - Client List
 - DHCP Clients
 - Network Map
8. Record at least three currently connected devices.

My Connected Devices

No	Device Name	MAC Address
.		

1.	[Enter actual device name]	[Enter actual MAC address]
2.	[Enter actual device name]	[Enter actual MAC address]
3.	[Enter actual device name]	[Enter actual MAC address]

Task 3: Comparison of Router, Modem and Switch

Feature	Router	Modem	Switch
Main Function	Connects different networks and routes packets between them.	Connects the local network to the ISP's access network.	Connects multiple devices within a LAN.
Used at Home/Office	Both home and office	Both home and office	Mainly office/LAN environments, but also useful at home for additional wired devices.
Assigns IP Addresses?	Yes, commonly through DHCP on home networks.	Usually No; the router or ISP equipment normally handles local DHCP.	Usually No for a basic Layer 2 switch.
Wi-Fi Support	Many modern routers provide Wi-Fi.	Usually no, although ISP devices may combine modem/router/Wi-Fi functions.	Usually no; a standard Ethernet switch provides wired connectivity.
OSI Layer	Mainly Layer 3	Depends on technology; acts	Mainly Layer 2

		as an access/physical-layer device rather than a normal LAN router	
Example Use	Connect laptops and phones to the Internet	Provide the Internet connection from the ISP	Connect multiple desktop PCs with Ethernet

Task 4: Small Gaming Café Network

Example Scenario

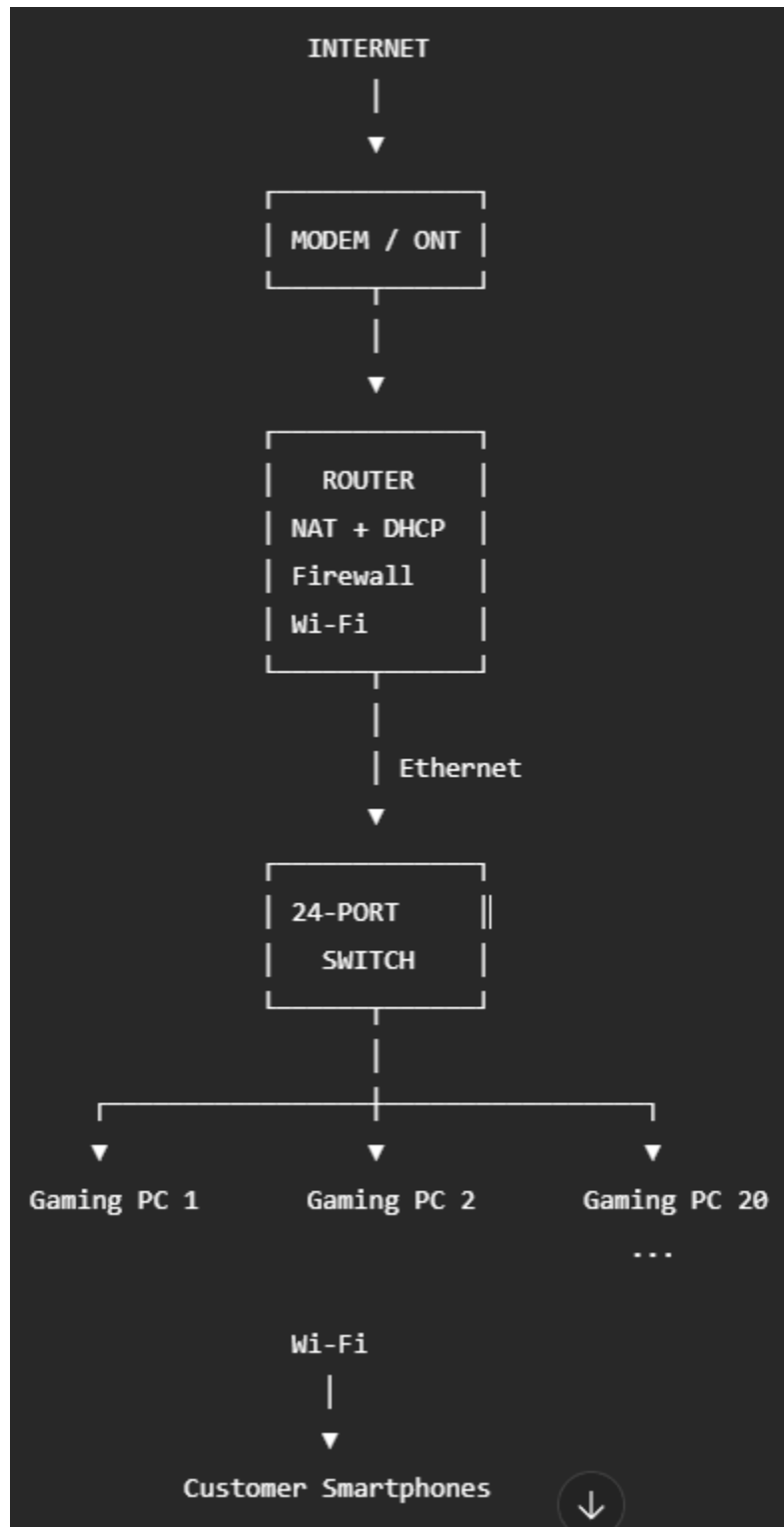
Assume the gaming café has:

- 20 gaming computers
- 1 café administrator computer
- Wi-Fi for customers' smartphones
- One Internet connection from an ISP

Required Devices

Device	Quantity	Reason
Modem/ONT	1	Connects the café network to the ISP's Internet service.
Router	1	Routes traffic between the café LAN and Internet and can provide DHCP, NAT, firewalling, and Wi-Fi.
24-port Gigabit switch	1	Provides enough Ethernet ports for the 20 gaming PCs plus the administrator PC and uplink to the router, with some spare ports.

Network Design



Why These Devices?

1 Modem/ONT:

One Internet connection is sufficient for the café, so one ISP termination device is required.

1 Router:

The router manages the connection between the local gaming network and the Internet. It can also provide DHCP, NAT, firewall protection, and wireless access.

1 24-port switch:

A 24-port switch is suitable for 20 gaming PCs because it provides enough ports while leaving a few spare ports for additional devices.